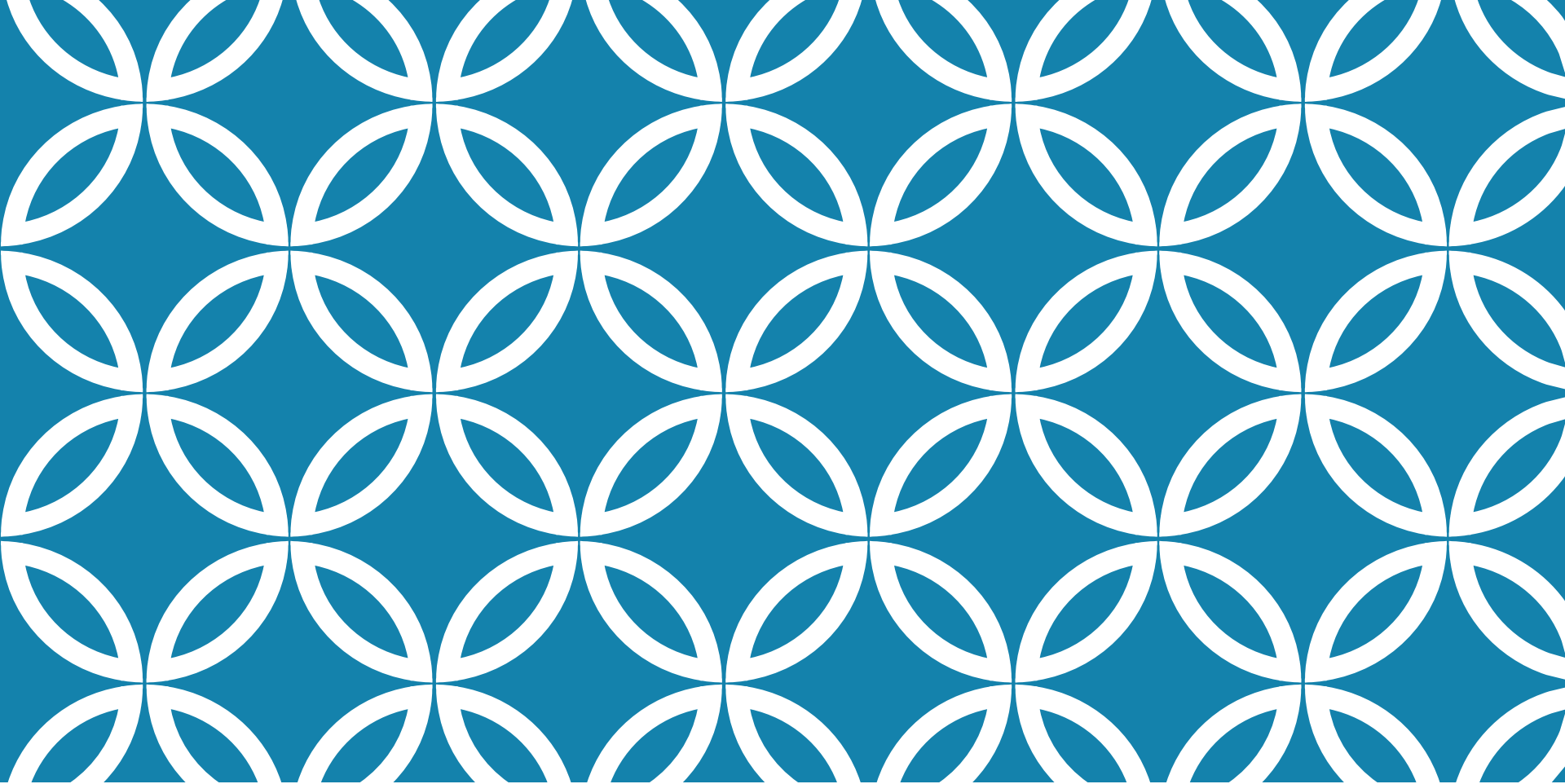




IT SOCIETY

INTRODUCTION

DATABASES

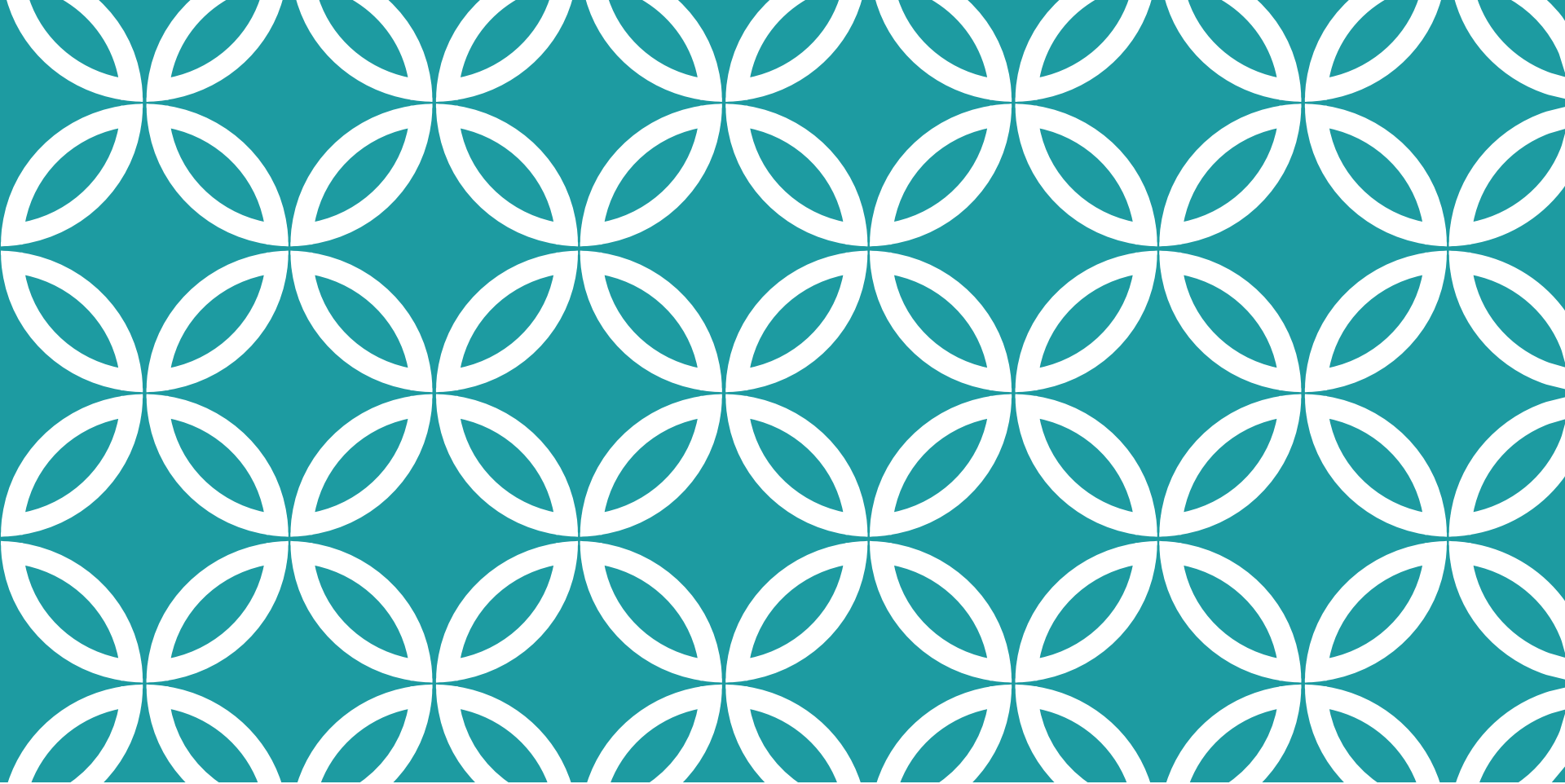
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RELATIONAL DATABASES





UNIONS

UNION,
UNION DISTINCT
& UNION ALL

INTRODUCTION

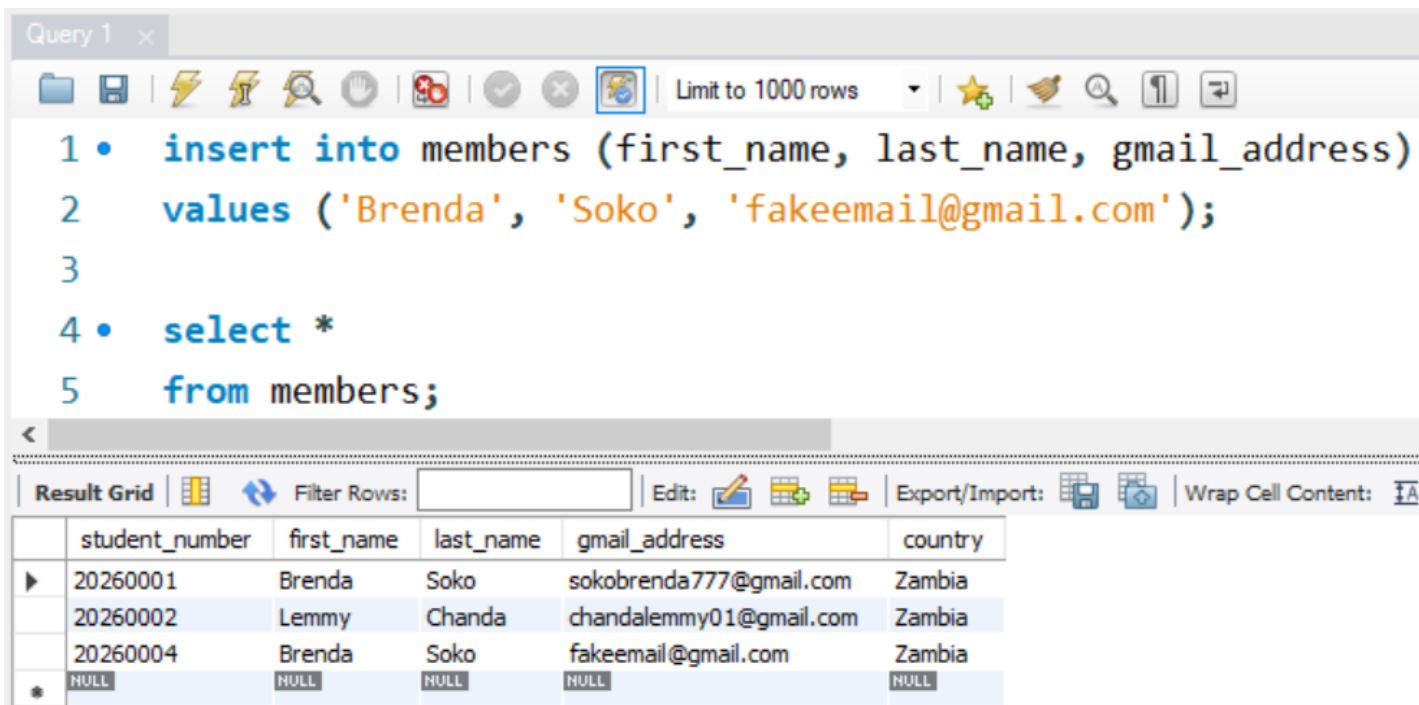
- The **UNION** operator is used to combine the result-set of two or more **SELECT** statements.
- Types of UNIONS include –
 1. UNION
 2. UNION DISTINCT
 3. UNION ALL

TYPES OF UNIONS

OPERATION	DESCRIPTION
UNION	Combines results, removes duplicates.
UNION DISTINCT	Explicit form of UNION.
UNION ALL	Combines results, keeps all rows.

21. UNION

- We need to make sure we have equal number of columns and for demonstration let's create some duplicate.



The screenshot shows a database query editor window titled "Query 1". The SQL code is as follows:

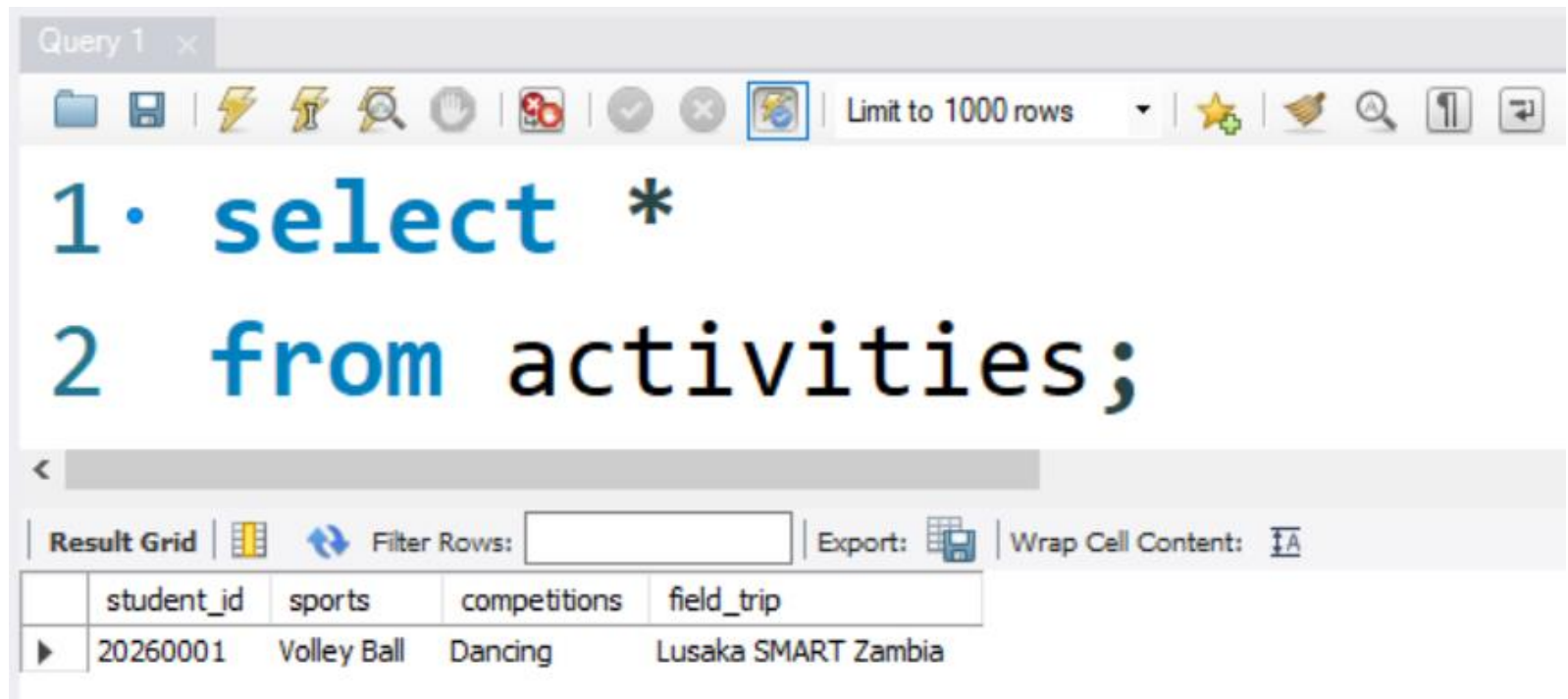
```
1 • insert into members (first_name, last_name, gmail_address)
2   values ('Brenda', 'Soko', 'fakeemail@gmail.com');
3
4 • select *
5   from members;
```

Below the code is a "Result Grid" showing the output of the query. It has 6 columns: student_number, first_name, last_name, gmail_address, and country. The data is as follows:

	student_number	first_name	last_name	gmail_address	country
▶	20260001	Brenda	Soko	sokobrenda777@gmail.com	Zambia
	20260002	Lemmy	Chanda	chandalemmmy01@gmail.com	Zambia
	20260004	Brenda	Soko	fakeemail@gmail.com	Zambia
*	NULL	NULL	NULL	NULL	NULL

21. UNION (CONTINUES...)

- Now, let's take a look at our activities table



The screenshot shows a database query editor interface. At the top, there's a tab labeled "Query 1". Below the tab is a toolbar with various icons for file operations, execution, and viewing. A dropdown menu shows "Limit to 1000 rows". The main area contains the following SQL query:

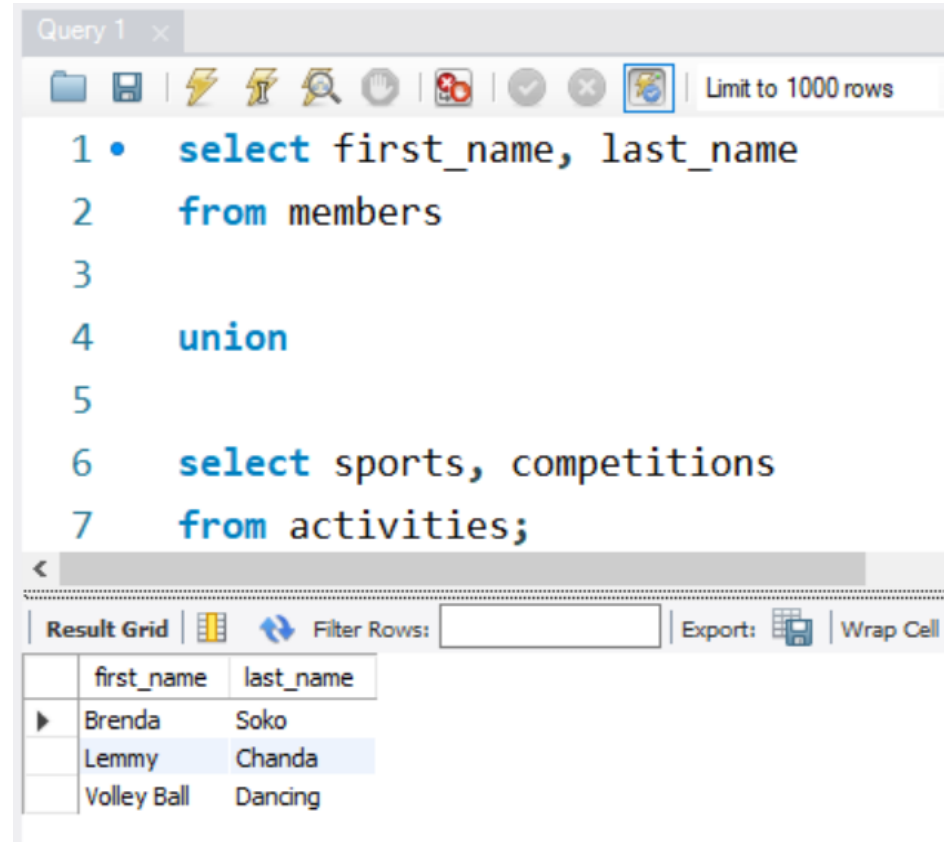
```
1. select *  
2 from activities;
```

Below the query editor, there's a "Result Grid" section. It includes a "Filter Rows" input field, an "Export" button, and a "Wrap Cell Content" checkbox. The results are displayed in a table with the following data:

	student_id	sports	competitions	field_trip
▶	20260001	Volley Ball	Dancing	Lusaka SMART Zambia

21. UNION (CONTINUES...)

- Now suppose we want to see the first and last name, and sports and competitions from our two tables.
- **NOTICE:** The **UNION** statement only prints **UNIQUE** or **DISTINCT** records.
- Recall Brenda appears twice.



The screenshot shows a database query editor window titled "Query 1". The query text is as follows:

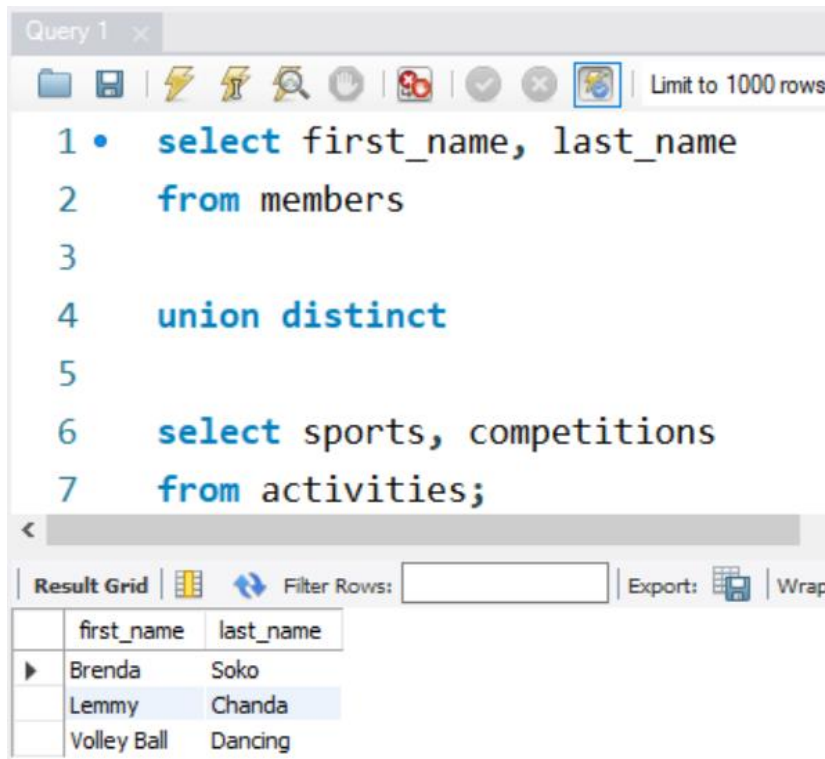
```
1 • select first_name, last_name
2   from members
3
4   union
5
6   select sports, competitions
7   from activities;
```

Below the query editor, the "Result Grid" is displayed, showing the results of the query. The grid has two columns: "first_name" and "last_name". The results are as follows:

	first_name	last_name
▶	Brenda	Soko
	Lemmy	Chanda
	Volley Ball	Dancing

22. UNION DISTINCT

- This is the same as UNION



The screenshot shows a database query editor window titled "Query 1". The query text is as follows:

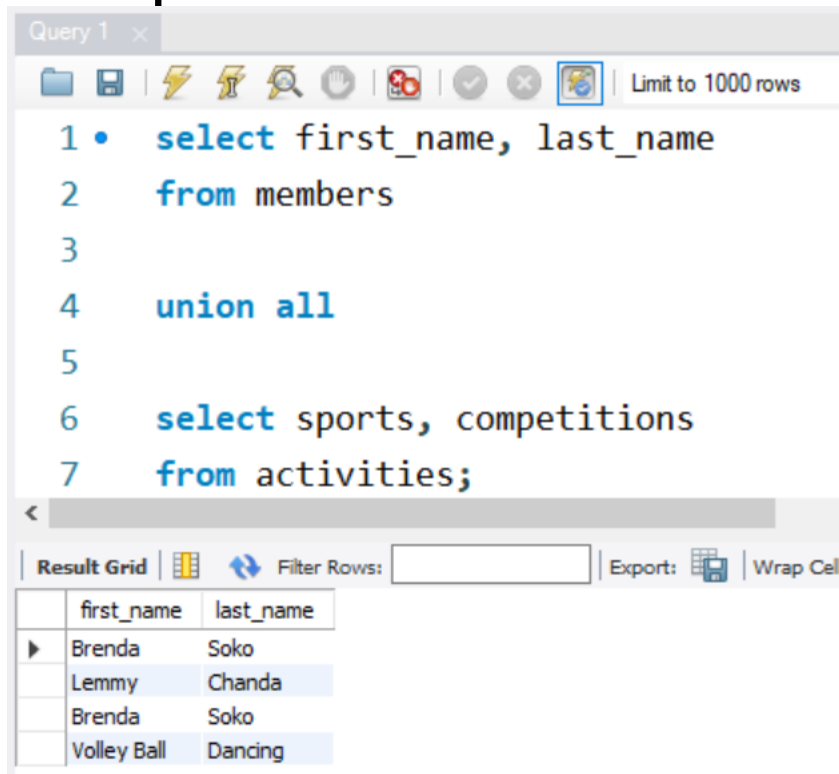
```
1 • select first_name, last_name
2   from members
3
4   union distinct
5
6   select sports, competitions
7   from activities;
```

Below the query editor, the "Result Grid" is displayed, showing the results of the query. The results are as follows:

	first_name	last_name
▶	Brenda	Soko
	Lemmy	Chanda
	Volley Ball	Dancing

23. UNION ALL

- With this all records will be displayed including duplicates.



The screenshot shows a database query editor window titled "Query 1". The query text is as follows:

```
1 • select first_name, last_name
2   from members
3
4   union all
5
6   select sports, competitions
7   from activities;
```

Below the query editor, the "Result Grid" is displayed, showing the results of the query. The results are as follows:

	first_name	last_name
▶	Brenda	Soko
	Lemmy	Chanda
	Brenda	Soko
	Volley Ball	Dancing

CONCLUSION

- At this point, we now have an understanding about some types of **UNIONS** in **MySQL**.
- In this handout, our focus was to understand the basic use of **UNION, UNION DISTINCT, and UNION ALL**.



UP NEXT... QUERYING DATA LANGUAGE (DQL)